

Assignment 1

Full-Stack

Web Development Assignment

1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Frontend Development:

Frontend is the part of a website or web application that users can see and interact with. It is built using HTML, CSS, and JavaScript.

Example: On Amazon, the product images, search bar, buttons, and menus are part of the frontend.

Backend Development:

Backend is the server-side part of a website. It processes requests, stores data, and communicates with the database.

Technologies: Python (Flask, Django), Node.js, Java, PHP.

Example: When a user logs into Amazon, the backend verifies the username and password from the database.

Full-Stack Development:

A full-stack developer works on both frontend and backend.

Example: A developer who creates both the Amazon shopping interface and the server that processes orders is a full-stack developer.

2. Create a simple diagram showing how the client-server model works in web architecture.

Client (Browser)

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HTTP Request

|

v

Web Server

|

HTML/CSS/JS Response

|

v

Client (Browser)

3. Describe how a browser requests and displays a web page from a web server.

1. User enters a website URL.
2. Browser sends an HTTP request to the web server.
3. The server processes the request.
4. The server sends HTML, CSS, JavaScript, and other files.
5. The browser renders the page.
6. The webpage becomes visible to the user.

4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

VS Code – Code editor

Web Browser – View and test websites

Git – Version control

GitHub – Store and share code

Node.js – Run JavaScript outside the browser

Live Server Extension – Automatically reloads web pages

Chrome DevTools – Debug HTML, CSS, and JavaScript

5. Explain what a web server is and give examples of commonly used servers.

A web server is software or hardware that receives requests from clients and sends web pages back to them.

Examples: Apache HTTP Server, Nginx, Microsoft IIS, LiteSpeed, Caddy.

6. Define the roles of a frontend developer, backend developer, and database administrator in a project.

Frontend Developer:

- Designs the user interface using HTML, CSS, and JavaScript.

Backend Developer:

- Develops server-side logic, authentication, APIs, and database connectivity.

Database Administrator (DBA):

- Manages databases, backups, security, and performance optimization.

7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

Steps:

1. Install Visual Studio Code.
2. Install extensions:
 - Live Server
 - HTML CSS Support
 - JavaScript (ES6) Snippets
 - Prettier

3. Open a project folder.

Note: The screenshot must be taken on your own computer after completing the setup.

8. Explain the difference between static and dynamic websites. Provide an example of each.

Static Website:

- Same content for every user.
- No database required.

Example: Portfolio website.

Dynamic Website:

- Content changes according to the user.
- Uses a database.

Example: Amazon, Facebook, YouTube.

9. Research and list five web browsers. Explain how rendering engines differ between them.

Google Chrome – Blink

Microsoft Edge – Blink

Mozilla Firefox – Gecko

Apple Safari – WebKit

Opera – Blink

Blink is used by Chromium-based browsers, Gecko is Mozilla's independent engine, and WebKit is Apple's rendering engine for Safari.

10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

User

|

Browser

|

HTTP Request

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Web Server

|

| +--> External APIs

|

+--> Database

|

HTTP Response

|

Browser

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